

INTERNATIONAL STANDARD

IEC
60191-6-6

First edition
2001-03

Mechanical standardization of semiconductor devices –

Part 6-6: General rules for the preparation of outline drawings of surface mounted semiconductor device packages – Design guide for fine pitch land grid array (FLGA)

Normalisation mécanique des dispositifs à semi-conducteurs –

*Partie 6-6:
Règles générales pour la préparation des dessins
d'encombrement des dispositifs à semi-conducteurs
pour montage en surface – Guide de conception
des dispositifs FLGA*



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International Electrotechnical Commission 3, rue de Varembé Geneva, Switzerland
Telefax: +41 22 919 0300 e-mail: inmail@iec.ch IEC web site <http://www.iec.ch>



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INTERNATIONAL ELECTROTECHNICAL COMMISSION

MECHANICAL STANDARDIZATION OF SEMICONDUCTOR DEVICES –

**Part 6-6: General rules for the preparation of outline drawings
of surface mounted semiconductor device packages –
Design guide for fine-pitch land grid array (FLGA)**

FOREWORD

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International Standard IEC 60191-6-6 has been prepared by subcommittee 47D: Mechanical standardization of semiconductor devices, of IEC technical committee 47: Semiconductor devices.

The text of this standard is based on the following documents:

FDIS	Report on voting
47D/404/FDIS	47D/421/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 3.

The committee has decided that the contents of this publication will remain unchanged until 2003. At this date, the publication will be

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

A bilingual version of this standard may be issued at a later date.

INTRODUCTION

The demand for area array style packages exists because of the multi-functions and high performance of electrical equipment. The objective of this design guide is to standardize outlines and to get interchangeability of FLGA packages. The terminal pitch and package outlines of these fine-pitch array packages are smaller than those of LGA packages.

MECHANICAL STANDARDIZATION OF SEMICONDUCTOR DEVICES –

Part 6-6: General rules for the preparation of outline drawings of surface mounted semiconductor device packages – Design guide for fine-pitch land grid array (FLGA)

1 Scope

This part of IEC 60191 provides common outline drawings and dimensions for all types of structures and composed materials of fine-pitch land grid array (hereinafter called FLGA) whose terminal pitch is less than, or equal to, 0,80 mm and whose package body outline is square.

2 Normative references

The following normative documents contain provisions which, through reference in this text, constitute provisions of this part of IEC 60191. For dated references, subsequent amendments to, or revisions of, any of these publications do not apply. However, parties to agreements based on this part of IEC 60191 are encouraged to investigate the possibility of applying the most recent editions of the normative documents indicated below. For undated references, the latest edition of the normative document referred to applies. Members of IEC and ISO maintain registers of currently valid International Standards.

IEC 60191 (all parts), *Mechanical standardization of semiconductor devices*

3 Definitions

For the purposes of this part of IEC 60191, the following definitions, as well as those given in the other parts of this standard, apply.

3.1

flanged type

type whose package body size (body length and width) consists of its own flange composed around the encapsulation or lid

3.2

type of real chip size

type whose package body size (body length and width) consists of an encapsulation around the real chip only

3.3

FLGA

packages with metal lands or metal bumps of which the terminal height is less than, or equal to, 100 μm , and whose terminal pitch is less than, or equal to, 0,80 mm, positioned in an array on the base plane of the package as external terminals

This package structure makes it possible to surface-mount the packages to the printed circuit board

3.4

material designation

FLGA packages are classified according to the following two material designations:

3.4.1

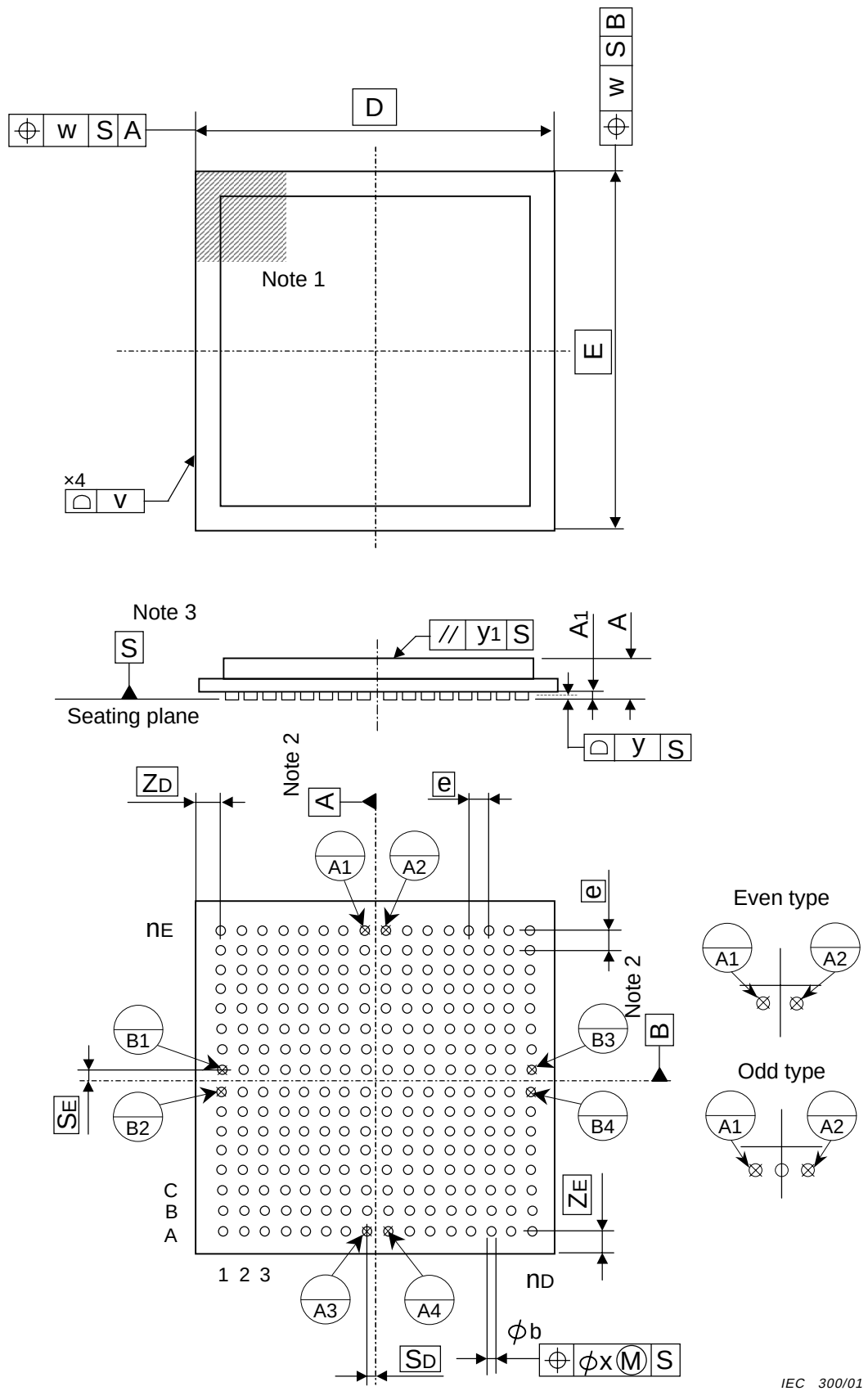
plastic type (P-FLGA)

plastic-type classification is assigned to packages which consist of resin substrate as interposer material (for example, glass-epoxy, poly-imid)

3.4.2

ceramic type (C-FLGA)

ceramic-type classification is assigned to packages which consist of ceramic substrate as interposer material



	Design guide for fine-pitch land grid array family	
	IEC 60191	

NOTE 1 Zone of a visible index on the top surface.

NOTE 2 Datum A and B are the axes defined by the terminal positions indicated with datum targets.

NOTE 3 Primary datum S and seating plane to be defined by the method of least squares of spherical crowns of terminals.

Table 1 – Group 1: Dimensions appropriate to mounting and interchangeability

Ref.	Limits to be observed			Recommended values for the dimensions mm	Note																																								
	Min.	Nom.	Max.																																										
n		X			1, 2																																								
nD nE		X X			1																																								
A			X	A max. = 1,20, 1,70, 2,00	Includes heat slug Includes package warpage and tilt																																								
A ₁			X	A ₁ max. = 0,10																																									
Øb	X	X	X	At ceramic FLGA (C-FLGA) <table><tr><td></td><td>Min.</td><td>Nom.</td><td>Max.</td></tr><tr><td>at e = 0,80</td><td>0,45</td><td>0,50</td><td>0,55</td></tr><tr><td>at e = 0,65</td><td>0,35</td><td>0,40</td><td>0,45</td></tr><tr><td>at e = 0,50</td><td>0,25</td><td>0,30</td><td>0,35</td></tr><tr><td>at e = 0,40</td><td>0,20</td><td>0,25</td><td>0,30</td></tr></table> At plastic FLGA (P-FLGA) <table><tr><td></td><td>Min.</td><td>Nom.</td><td>Max.</td></tr><tr><td>at e = 0,80</td><td>0,35</td><td>0,40</td><td>0,45</td></tr><tr><td>at e = 0,65</td><td>0,28</td><td>0,33</td><td>0,38</td></tr><tr><td>at e = 0,50</td><td>0,20</td><td>0,25</td><td>0,30</td></tr><tr><td>at e = 0,40</td><td>0,15</td><td>0,20</td><td>0,25</td></tr></table>		Min.	Nom.	Max.	at e = 0,80	0,45	0,50	0,55	at e = 0,65	0,35	0,40	0,45	at e = 0,50	0,25	0,30	0,35	at e = 0,40	0,20	0,25	0,30		Min.	Nom.	Max.	at e = 0,80	0,35	0,40	0,45	at e = 0,65	0,28	0,33	0,38	at e = 0,50	0,20	0,25	0,30	at e = 0,40	0,15	0,20	0,25	
	Min.	Nom.	Max.																																										
at e = 0,80	0,45	0,50	0,55																																										
at e = 0,65	0,35	0,40	0,45																																										
at e = 0,50	0,25	0,30	0,35																																										
at e = 0,40	0,20	0,25	0,30																																										
	Min.	Nom.	Max.																																										
at e = 0,80	0,35	0,40	0,45																																										
at e = 0,65	0,28	0,33	0,38																																										
at e = 0,50	0,20	0,25	0,30																																										
at e = 0,40	0,15	0,20	0,25																																										
		X		At flanged type D = 4,0, 5,0, 6,0, 7,0, 8,0, 9,0, 10,0, 11,0, 12,0, 13,0, 14,0, 15,0, 16,0, 17,0, 18,0, 19,0, 20,0, 21,0 At type of real chip size D = from 3,1 to 21,0	Dimension range shows nominal value																																								
		X		At flanged type E = 4,0, 5,0, 6,0, 7,0, 8,0, 9,0, 10,0, 11,0, 12,0, 13,0, 14,0, 15,0, 16,0, 17,0, 18,0, 19,0, 20,0, 21,0 At type of real chip size E = from 3,1 to 21,0	Dimension range shows nominal value																																								
		X		e = 0,80, 0,65, 0,50, 0,40																																									
v			X	v = 0,15	Includes burrs																																								
w			X	at e = 0,80 w = 0,20 at e = 0,65 w = 0,20 at e = 0,50 w = 0,20 at e = 0,40 w = 0,15																																									
x			X	at e = 0,80 x = 0,08 at e = 0,65 x = 0,08 at e = 0,50 x = 0,05 at e = 0,40 x = 0,05																																									

Table 1 – (continued)

Ref.	Limits to be observed			Recommended values for the dimensions mm	Note
	Min.	Nom.	Max.		
y			X	at e = 0,80 y = 0,10 at e = 0,65 y = 0,10 at e = 0,50 y = 0,08 at e = 0,40 y = 0,08	
y ₁			X	y ₁ = 0,2	
NOTE 1 The values stipulated by the mathematical expression should be applied to the individual overall dimensional standards.					
NOTE 2 Symbol n refers to the total number of terminal positions.					

Table 2 – Group 2: Dimensions appropriate to mounting and gauging

Ref.	Limits to be observed			Recommended values for the dimensions mm	Note
	Min.	Nom.	Max.		
Øb2			X	At ceramic FLGA (C-FLGA) at e = 0,80 b2 = 0,63 at e = 0,65 b2 = 0,53 at e = 0,50 b2 = 0,40 at e = 0,40 b2 = 0,35 At plastic FLGA (P-FLGA) at e = 0,80 b2 = 0,53 at e = 0,65 b2 = 0,46 at e = 0,50 b2 = 0,35 at e = 0,40 b2 = 0,30	
		X		e = 0,80, 0,65, 0,50, 0,40	
		X		eD = e x (nD – 1)	1 ^a
		X		eE = e x (nE – 1)	
^a See note 1 of table 1.					

Table 3 – Group 3: Dimensions appropriate to automated handling

Ref.	Limits to be observed			Recommended values for the dimensions mm	Note
	Min.	Nom.	Max.		
A			X	A max. = 1,20, 1,70, 2,00	Includes heat slug Includes package warpage and tilt
		X		D/E = 4,0, 5,0, 6,0, 7,0, 8,0, 9,0, 10,0, 11,0, 12,0, 13,0, 14,0, 15,0, 16,0, 17,0, 18,0, 19,0, 20,0, 21,0	
		X			
y ₁			X	y ₁ = 0,2	

Table 4 – Group 4: Dimensions for information only

Ref.	Limits to be observed			Recommended values for the dimensions mm	Note
	Min.	Nom.	Max.		
		X		$ZD = (D - e \times (nD - 1))/2$	1 ^a
		X		$ZE = (E - e \times (nE - 1))/2$	
^a See note 1 of table 1.					

Table 5 – Dimensions and ball matrix

e = 0,80	Maximum matrix row family					Maximum matrix – 1 row family					Maximum matrix – 2 row family				
E × D	nD	nE	n	$\overline{ZD}$	$\overline{ZE}$	nD	nE	n	$\overline{ZD}$	$\overline{ZE}$	nD	nE	n	$\overline{ZD}$	$\overline{ZE}$
4 × 4	–	–	–	–	–	4	4	16	0,80	0,80	3	3	9	1,20	1,20
5 × 5	6	6	36	0,50	0,50	5	5	25	0,90	0,90	4	4	16	1,30	1,30
6 × 6	–	–	–	–	–	7	7	49	0,60	0,60	6	6	36	1,00	1,00
7 × 7	–	–	–	–	–	8	8	64	0,70	0,70	7	7	49	1,10	1,10
8 × 8	–	–	–	–	–	9	9	81	0,80	0,80	8	8	64	1,20	1,20
9 × 9	11	11	121	0,50	0,50	10	10	100	0,90	0,90	9	9	81	1,30	1,30
10 × 10	–	–	–	–	–	12	12	144	0,60	0,60	11	11	121	1,00	1,00
11 × 11	–	–	–	–	–	13	13	169	0,70	0,70	12	12	144	1,10	1,10
12 × 12	–	–	–	–	–	14	14	196	0,80	0,80	13	13	169	1,20	1,20
13 × 13	16	16	256	0,50	0,50	15	15	225	0,90	0,90	14	14	196	1,30	1,30
14 × 14	–	–	–	–	–	17	17	289	0,60	0,60	16	16	256	1,00	1,00
15 × 15	–	–	–	–	–	18	18	324	0,70	0,70	17	17	289	1,10	1,10
16 × 16	–	–	–	–	–	19	19	361	0,80	0,80	18	18	324	1,20	1,20
17 × 17	21	21	441	0,50	0,50	20	20	400	0,90	0,90	19	19	361	1,30	1,30
18 × 18	–	–	–	–	–	22	22	484	0,60	0,60	21	21	441	1,00	1,00
19 × 19	–	–	–	–	–	23	23	529	0,70	0,70	22	22	484	1,10	1,10
20 × 20	–	–	–	–	–	24	24	576	0,80	0,80	23	23	529	1,20	1,20
21 × 21	26	26	676	0,50	0,50	25	25	625	0,90	0,90	24	24	576	1,30	1,30
e = 0,65	Maximum matrix row family					Maximum matrix – 1 row family					Maximum matrix – 2 row family				
E × D	nD	nE	n	$\overline{ZD}$	$\overline{ZE}$	nD	nE	n	$\overline{ZD}$	$\overline{ZE}$	nD	nE	n	$\overline{ZD}$	$\overline{ZE}$
4 × 4	–	–	–	–	–	5	5	25	0,700	0,700	4	4	16	1,025	1,025
5 × 5	–	–	–	–	–	7	7	49	0,550	0,550	6	6	36	0,875	0,875
6 × 6	–	–	–	–	–	8	8	64	0,725	0,725	7	7	49	1,050	1,050
7 × 7	–	–	–	–	–	10	10	100	0,575	0,575	9	9	81	0,900	0,900
8 × 8	–	–	–	–	–	11	11	121	0,750	0,750	10	10	100	1,075	1,075
9 × 9	–	–	–	–	–	13	13	169	0,600	0,600	12	12	144	0,925	0,925
10 × 10	15	15	225	0,450	0,450	14	14	196	0,775	0,775	13	13	169	1,100	1,100
11 × 11	–	–	–	–	–	16	16	256	0,625	0,625	15	15	225	0,950	0,950
12 × 12	18	18	324	0,475	0,475	17	17	289	0,800	0,800	16	16	256	1,125	1,125
13 × 13	–	–	–	–	–	19	19	361	0,650	0,650	18	18	324	0,975	0,975
14 × 14	21	21	441	0,500	0,500	20	20	400	0,825	0,825	19	19	361	1,150	1,150
15 × 15	–	–	–	–	–	22	22	484	0,675	0,675	21	21	441	1,000	1,000
16 × 16	24	24	576	0,525	0,525	23	23	529	0,850	0,850	22	22	484	1,175	1,175
17 × 17	–	–	–	–	–	25	25	625	0,700	0,700	24	24	576	1,025	1,025
18 × 18	–	–	–	–	–	27	27	729	0,550	0,550	26	26	676	0,875	0,875
19 × 19	–	–	–	–	–	28	28	784	0,725	0,725	27	27	729	1,050	1,050
20 × 20	–	–	–	–	–	30	30	900	0,575	0,575	29	29	841	0,900	0,900
21 × 21	–	–	–	–	–	31	31	961	0,750	0,750	30	30	900	1,075	1,075

Table 5 – (continued)

e = 0,50	Maximum matrix row family					Maximum matrix – 1 row family					Maximum matrix – 2 row family				
E × D	nD	nE	n	ZD	ZE	nD	nE	n	ZD	ZE	nD	nE	n	ZD	ZE
4 × 4	–	–	–	–	–	7	7	49	0,50	0,50	6	6	36	0,75	0,75
5 × 5	–	–	–	–	–	9	9	81	0,50	0,50	8	8	64	0,75	0,75
6 × 6	–	–	–	–	–	11	11	121	0,50	0,50	10	10	100	0,75	0,75
7 × 7	–	–	–	–	–	13	13	169	0,50	0,50	12	12	144	0,75	0,75
8 × 8	–	–	–	–	–	15	15	225	0,50	0,50	14	14	196	0,75	0,75
9 × 9	–	–	–	–	–	17	17	289	0,50	0,50	16	16	256	0,75	0,75
10 × 10	–	–	–	–	–	19	19	361	0,50	0,50	18	18	324	0,75	0,75
11 × 11	–	–	–	–	–	21	21	441	0,50	0,50	20	20	400	0,75	0,75
12 × 12	–	–	–	–	–	23	23	529	0,50	0,50	22	22	484	0,75	0,75
13 × 13	–	–	–	–	–	25	25	625	0,50	0,50	24	24	576	0,75	0,75
14 × 14	–	–	–	–	–	27	27	729	0,50	0,50	26	26	676	0,75	0,75
15 × 15	–	–	–	–	–	29	29	841	0,50	0,50	28	28	784	0,75	0,75
16 × 16	–	–	–	–	–	31	31	961	0,50	0,50	30	30	900	0,75	0,75
17 × 17	–	–	–	–	–	33	33	1089	0,50	0,50	32	32	1024	0,75	0,75
18 × 18	–	–	–	–	–	35	35	1225	0,50	0,50	34	34	1156	0,75	0,75
19 × 19	–	–	–	–	–	37	37	1369	0,50	0,50	36	36	1296	0,75	0,75
20 × 20	–	–	–	–	–	39	39	1521	0,50	0,50	38	38	1444	0,75	0,75
21 × 21	–	–	–	–	–	41	41	1681	0,50	0,50	40	40	1600	0,75	0,75
e = 0,40	Maximum matrix row family					Maximum matrix – 1 row family					Maximum matrix – 2 row family				
E × D	nD	nE	n	ZD	ZE	nD	nE	n	ZD	ZE	nD	nE	n	ZD	ZE
4 × 4	9	9	81	0,40	0,40	8	8	64	0,60	0,60	7	7	49	0,80	0,80
5 × 5	–	–	–	–	–	11	11	121	0,50	0,50	10	10	100	0,70	0,70
6 × 6	14	14	196	0,40	0,40	13	13	169	0,60	0,60	12	12	144	0,80	0,80
7 × 7	–	–	–	–	–	16	16	256	0,50	0,50	15	15	225	0,70	0,70
8 × 8	19	19	361	0,40	0,40	18	18	324	0,60	0,60	17	17	289	0,80	0,80
9 × 9	–	–	–	–	–	21	21	441	0,50	0,50	20	20	400	0,70	0,70
10 × 10	24	24	576	0,40	0,40	23	23	529	0,60	0,60	22	22	484	0,80	0,80
11 × 11	–	–	–	–	–	26	26	676	0,50	0,50	25	25	625	0,70	0,70
12 × 12	29	29	841	0,40	0,40	28	28	784	0,60	0,60	27	27	729	0,80	0,80
13 × 13	–	–	–	–	–	31	31	961	0,50	0,50	30	30	900	0,70	0,70
14 × 14	34	34	1156	0,40	0,40	33	33	1089	0,60	0,60	32	32	1024	0,80	0,80
15 × 15	–	–	–	–	–	36	36	1296	0,50	0,50	35	35	1225	0,70	0,70
16 × 16	39	39	1521	0,40	0,40	38	38	1444	0,60	0,60	37	37	1369	0,80	0,80
17 × 17	–	–	–	–	–	41	41	1681	0,50	0,50	40	40	1600	0,70	0,70
18 × 18	44	44	1936	0,40	0,40	43	43	1849	0,60	0,60	42	42	1764	0,80	0,80
19 × 19	–	–	–	–	–	46	46	2116	0,50	0,50	45	45	2025	0,70	0,70
20 × 20	49	49	2401	0,40	0,40	48	48	2304	0,60	0,60	47	47	2209	0,80	0,80
21 × 21	–	–	–	–	–	51	51	2601	0,50	0,50	50	50	2500	0,70	0,70
NOTE The relations among package body size, maximum number of terminals, maximum number of matrices, terminal pitch, and package over-hang are shown in this table. (Depopulated matrices and numbers of terminals are also involved.)															



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Fax to: **IEC/CSC** at +41 22 919 03 00

Thank you for your contribution to the standards-making process.

A Prioritaire

Nicht frankieren
Ne pas affranchir



Non affrancare
No stamp required

RÉPONSE PAYÉE

SUISSE

Customer Service Centre (CSC)
International Electrotechnical Commission
3, rue de Varembe
1211 GENEVA 20
Switzerland



Q1 Please report on **ONE STANDARD** and **ONE STANDARD ONLY**. Enter the exact number of the standard: (e.g. 60601-1-1)

.....

Q2 Please tell us in what capacity(ies) you bought the standard (tick all that apply). I am the/a:

- purchasing agent ☐
 librarian ☐
 researcher ☐
 design engineer ☐
 safety engineer ☐
 testing engineer ☐
 marketing specialist ☐
 other.....

Q3 I work for/in/as a:
(tick all that apply)

- manufacturing ☐
 consultant ☐
 government ☐
 test/certification facility ☐
 public utility ☐
 education ☐
 military ☐
 other.....

Q4 This standard will be used for:
(tick all that apply)

- general reference ☐
 product research ☐
 product design/development ☐
 specifications ☐
 tenders ☐
 quality assessment ☐
 certification ☐
 technical documentation ☐
 thesis ☐
 manufacturing ☐
 other.....

Q5 This standard meets my needs:
(tick one)

- not at all ☐
 nearly ☐
 fairly well ☐
 exactly ☐

Q6 If you ticked NOT AT ALL in Question 5 the reason is: (tick all that apply)

- standard is out of date ☐
 standard is incomplete ☐
 standard is too academic ☐
 standard is too superficial ☐
 title is misleading ☐
 I made the wrong choice ☐
 other

Q7 Please assess the standard in the following categories, using the numbers:

- (1) unacceptable,
 (2) below average,
 (3) average,
 (4) above average,
 (5) exceptional,
 (6) not applicable

- timeliness.....
 quality of writing.....
 technical contents.....
 logic of arrangement of contents
 tables, charts, graphs, figures.....
 other

Q8 I read/use the: (tick one)

- French text only ☐
 English text only ☐
 both English and French texts ☐

Q9 Please share any comment on any aspect of the IEC that you would like us to know:

.....



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